Roll No. : ....

MID SEMESTER EXAMINATION SEPTEMBER 2025

Name of the Program: **B.Tech. (CSE-all kind of specializations only)** Semester: V

Name of the Course: **Computer Networks** Course Code: **TCS-511**

Time: 90 Minutes **Maximum Marks: 50**

Note:

i. Answer all the questions by choosing any one of the sub questions
ii. Each questions carries 10 marks

Q1	(10 Marks)	
(a)	A user sends an email using SMTP. Trace the journey of this message through the layers of OSI reference and TCP/IP models until it reaches the recipient.	CO3
	OR	
(b)	Consider a network with six nodes (A, B, C, D, E, F), where each node wants to establish a dedicated circuit with every other node for data transmission. The network utilizes circuit switching, and each circuit requires 128 Kbps. Assume the setup time for each circuit is 15 seconds. i. Calculate the total number of circuits needed for full connectivity. ii. Determine the total setup time required for all circuits to be established.	CO4
Q2	(10 Marks)	C02
(a)	Explain the working of the Go-Back-N (GBN) and Selective Repeat protocol. Illustrate with a diagram showing the sender and receiver windows.	
	OR	
(b)	An HTTP request is sent from a client to a server over a network with an average propagation delay of 10 ms, queuing delay of 5 ms, processing delay of 2 ms at each router, and transmission delay of 1 ms per kilobyte. If the request size is 10 KB, what is the end-to-end delay?	CO5
Q3	(10 Marks)	
(a)	Explain the working of Non-Persistent HTTP connections. Elaborate how TCP connections are established and closed for each object requested?	CO1
	OR	
(b)	Explain the sequence of SMTP request and response messages exchanged between a client and mail server when sending a simple email. Also illustrate the how are email is retrieved by the recipient.	CO2
Q4	(10 Marks)	CO2
(a)	Compare Iterative DNS queries and Recursive DNS queries. Outline the steps involved in resolving the domain name www.example.com into an IP address, assuming no caching is involved.	
	OR	
(b)	A small business wants to connect to the Internet with reliable and fast access. Which type of ISP connection would you recommend (Tier 1, Tier 2, or Tier 3), and why?	CO3
Q5	(10 Marks)	CO2
(a)	Describe the strategies utilized by a BitTorrent client lacking any data chunks to initiate its download within a pre-existing swarm.	
	OR	
(b)	Why is TCP considered reliable and connection-oriented, while UDP is unreliable and connectionless? List at least three applications that use TCP and three that use UDP.	CO1